

PROJECT 5 – A/B TESTING PRODUCT ANALYTICS

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Tools Used: SQL | Power BI | DAX | Data Cleaning

DAX MEASURES USED

Total Users

Purpose: Total users included in the experiment.

```
Total User =  
DISTINCTCOUNT(ab_testing_clean[user_id])
```

Total Buyers

Purpose: Total users who completed a purchase.

```
Total Buyers =  
CALCULATE(  
    DISTINCTCOUNT(ab_testing_clean[user_id]),  
    ab_testing_clean[converted] = 1  
)
```

Total Revenue

Purpose: Total revenue generated during the experiment.

```
Total Amount =  
SUM(ab_testing_clean[revenue])
```

Conversion Rate

Purpose: Percentage of users who converted.

```

Conversion Rate =
DIVIDE (
    SUM(ab_testing_clean[converted]),
    DISTINCTCOUNT(ab_testing_clean[user_id]),
    0
)

```

ARPU

Purpose: Average Revenue Per User.

```

ARPU =
ROUND (
    DIVIDE([Total Amount],[Total User],0),
    2
)

```

Revenue Per Buyer

Purpose: Average revenue generated per converted customer.

```

Revenue By Buyers =
ROUND (
    DIVIDE([Total Amount],[Total Buyers],0),
    2
)

```

Conversion Lift

Purpose: Measures improvement of Test Variant over Control Variant.

```

Conversion Lift =
VAR Controll1 =
    CALCULATE([Conversion Rate],
        ab_testing_clean[variant] = "Control")

VAR Test2 =
    CALCULATE([Conversion Rate],
        ab_testing_clean[variant] = "Test")

RETURN
ROUND (

```

```
DIVIDE(Test2 - Controll1,
Controll1,0),
2)
```

Winning Variant

Purpose: Identifies the better-performing variant.

```
Winning Variant =
IF(
    [Conversion Lift] > 0,
    "Test",
    "Control"
)
```

BUSINESS KPIs COVERED

- ✓ Total Users
 - ✓ Total Buyers
 - ✓ Total Revenue
 - ✓ Conversion Rate
 - ✓ Conversion Lift
 - ✓ ARPU
 - ✓ Revenue Per Buyer
 - ✓ Winning Variant
-

PROJECT OUTCOME

- Test Conversion Rate: **14.06%**
- Control Conversion Rate: **9.43%**
- Conversion Lift: **49%**
- Test Revenue: **₹16.76K**
- Control Revenue: **₹10.48K**
- Winning Variant: **Test Variant**

Final Recommendation

Launch the **Test Variant** because it achieved higher conversion rates and generated significantly more revenue than the Control Variant.